

MACSE622 – DATA ENGINEERING AND VISUALIZATION LAB

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Programme	MTech CSE (Big Data Analytics)	Slot	L33 + L34
Course	Data Engineering and Visualization Lab	Semester	1 st Semester
Date	08-12-2025	Exp No	4

Ex No: 4. File Transfer Automation using Apache NiFi

Aim: To download, install, and configure Apache NiFi, and demonstrate automated file transfer between directories using GetFile and PutFile processors.

Software Requirements

1. Operating System: Windows
2. Java Development Kit (JDK): Version 21
3. Apache NiFi: Version 2.7.2

Implementation:

1. Downloading and Installing Apache NiFi:
 - a. Apache NiFi version 2.7.2 was downloaded from the official Apache NiFi website.
 - b. After downloading, the compressed archive (nifi-2.7.2-bin.zip) was extracted to the directory: **C:\nifi-2.7.2**
 - c. Apache NiFi is a powerful dataflow orchestration tool used for designing, automating, and monitoring data movement in real time. It supports drag-and-drop dataflow creation, helping users visually connect systems for data ingestion, transformation, and delivery.
2. Configuring the JAVA_HOME Environment Variable:
 - a. Since NiFi requires Java to run, the JAVA_HOME environment variable was configured to point to the JDK installation.
 - b. In PowerShell, the following commands were executed from the NiFi bin directory:
 - i. `$env:JAVA_HOME="C:\Program Files\Java\jdk-21\"`
 - ii. `$env:PATH="$env:JAVA_HOME\bin;$env:PATH"`

- c. This ensures NiFi can locate the Java runtime that it depends on for execution. As a verification step, check Java installation using: **java -version**

```
Windows PowerShell
NIFI_HOME=C:\nifi-2.7.2

PS C:\nifi-2.7.2\bin> .\nifi.cmd stop
JAVA_HOME not defined: using PATH for java command
JAVA_HOME=C:\Program Files\Java\jdk-21\bin
NIFI_HOME=C:\nifi-2.7.2

2026-01-12 11:35:34,540 INFO [main] org.apache.nifi.bootstrap.Command Application Process not running
PS C:\nifi-2.7.2\bin> .\nifi.cmd start
JAVA_HOME not defined: using PATH for java command
JAVA_HOME=C:\Program Files\Java\jdk-21\bin
NIFI_HOME=C:\nifi-2.7.2

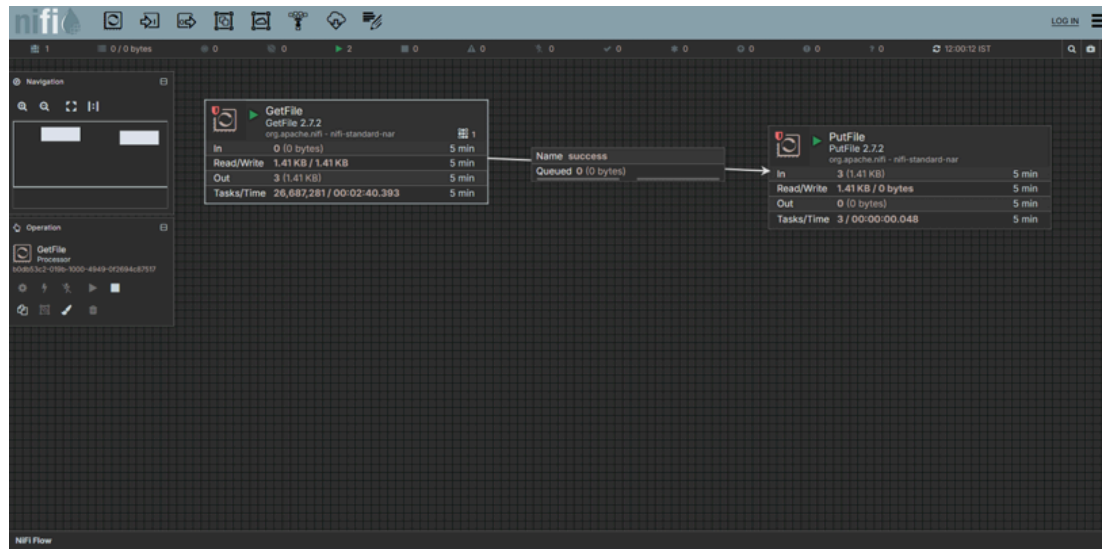
PS C:\nifi-2.7.2\bin> $env:JAVA_HOME="C:\Program Files\Java\jdk-21\"
PS C:\nifi-2.7.2\bin> $env:PATH="$env:JAVA_HOME\bin;$env:PATH"
PS C:\nifi-2.7.2\bin> .\nifi.cmd start
JAVA_HOME=C:\Program Files\Java\jdk-21\
NIFI_HOME=C:\nifi-2.7.2

PS C:\nifi-2.7.2\bin> .\nifi.cmd stop
JAVA_HOME=C:\Program Files\Java\jdk-21\
NIFI_HOME=C:\nifi-2.7.2

2026-01-12 11:38:02,071 INFO [main] org.apache.nifi.bootstrap.Command Bootstrap Process [16908] termination requested
2026-01-12 11:38:02,076 INFO [main] org.apache.nifi.bootstrap.Command Bootstrap Process [16908] termination completed
2026-01-12 11:38:02,076 INFO [main] org.apache.nifi.bootstrap.Command Application Process [15636] termination requested
2026-01-12 11:38:02,076 INFO [main] org.apache.nifi.bootstrap.Command Application Process [15636] termination completed
PS C:\nifi-2.7.2\bin> .\nifi.cmd start
JAVA_HOME=C:\Program Files\Java\jdk-21\
NIFI_HOME=C:\nifi-2.7.2
```

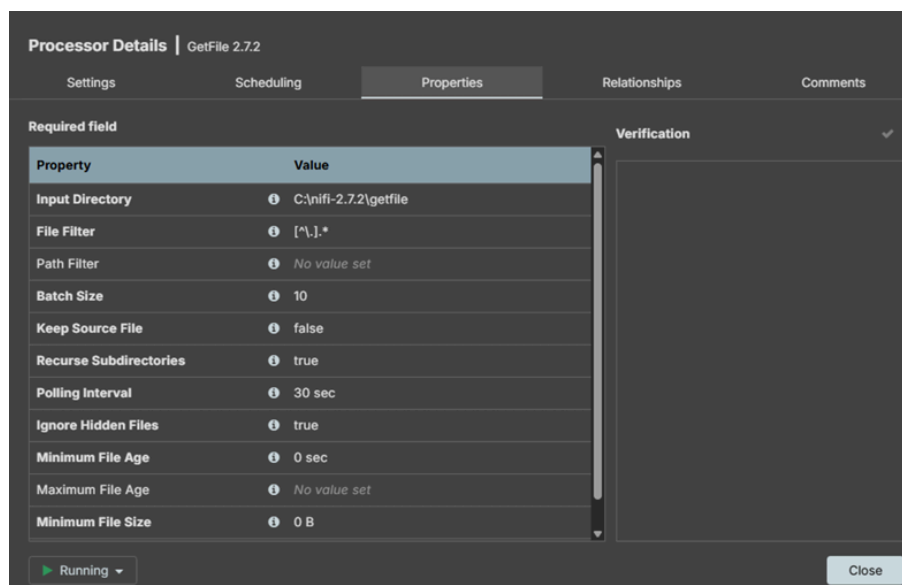
- 3. Starting Apache NiFi:
 - a. To start Apache NiFi, open PowerShell, go to bin directory, and run:
 - i. PS C:\nifi-2.7.2\bin> **.\nifi.cmd start**
 - b. Once started, NiFi displays messages showing: The detected JAVA_HOME path and The NIFI_HOME directory
 - c. To stop NiFi later, use: **.\nifi.cmd stop**
- 4. Accessing the NiFi Web Interface:
 - a. After the service starts, open browser and visit: <http://localhost:8080/nifi>
 - b. This launches the NiFi Web UI (User Interface), where we can build and monitor dataflows. The dashboard provides a canvas to drag processors, connect them with relationships, and configure data pipelines visually.
- 5. Creating a File Transfer Dataflow:
 - a. To demonstrate basic file movement, a simple dataflow using the GetFile and PutFile processors was created.
 - b. GetFile Processor: To read files from a specified input folder. It continuously monitors a source directory and fetches files as soon as they appear.
 - i. Input Directory: Folder path where files will be read.
 - ii. Keep Source File: False (so the source files can be deleted or moved after processing).
 - iii. File Filter: Specify patterns like *.txt to filter certain file types.
 - c. PutFile Processor: To write incoming files to a desired output directory. It saves data received from the GetFile processor to another folder.
 - i. Directory: Destination folder path.
 - ii. Conflict Resolution Strategy: Overwrite or fail if the file exists.

- iii. Create Missing Directories: True (if target folder doesn't exist).
- d. After configuration: Draw a connection between GetFile and PutFile processors. Click Start to schedule both processors.



6. Execution and Verification

- a. Once both processors are running:
 - i. Place any file (e.g. .txt or .csv file) into the source directory configured in GetFile.
 - ii. NiFi automatically detects and processes the file.
 - iii. The file is transferred to the destination directory configured in PutFile.
- b. We can monitor the workflow from the NiFi UI, each processor shows real-time stats like “FlowFiles In”, “Out”, and “Transferred”.
- c. If files successfully appear in the destination directory, the dataflow has executed correctly.



Processor Details | PutFile 2.7.2

Settings

Scheduling

Properties

Relationships

Comments

Required field

Property	Value
Directory	ⓘ C:\nifi-2.7.2\putfile
Conflict Resolution Strategy	ⓘ fail
Create Missing Directories	ⓘ true
Maximum File Count	ⓘ No value set
Last Modified Time	ⓘ No value set
Permissions	ⓘ No value set
Owner	ⓘ No value set
Group	ⓘ No value set

Verification

✓

▶ Running ▼

Close

Processor Details | PutFile 2.7.2

Settings

Scheduling

Properties

Relationships

Comments

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate
☐ retry

Files that could not be written to the output directory for some reason are transferred to this relationship

success

☒ terminate
☐ retry

Files that have been successfully written to the output directory are transferred to this relationship

▶ Running ▼

Close

Result: The file transfer workflow using GetFile and PutFile processors was successfully implemented. NiFi automatically moved files from the input folder to the output folder without manual intervention. Apache NiFi was successfully downloaded, installed, and configured on a Windows system with JDK 21. Through this experiment, the fundamental working of NiFi's processor-based architecture was demonstrated. A basic dataflow using GetFile and PutFile showcased NiFi's capability to automate file movement, enabling seamless data ingestion and delivery, an essential foundation for more advanced data pipeline automation tasks.

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